

RadeonShift AI

Accelerating the MI300X Hardware Migration via Generative AI

The Problem

Enterprise AI workloads are facing massive bottlenecks due to a reliance on legacy NVIDIA CUDA codebases. Manually migrating a 50,000-line CUDA codebase to ROCm takes 6+ months of specialized engineering time.

The Solution

RadeonShift AI is a deterministic MoA (Mixture of Agents) pipeline that translates CUDA to ROCm instantly. By leveraging `hipify-perl` for deterministic translation and Fireworks AI's `deepseek-v4-flash` for high-level architectural optimization, it removes manual engineering bottlenecks.

Market & Opportunity

- TAM: \$25B+ (Global AI Hardware & Software Services Market)
- SAM: \$8B (Enterprise AI Teams)
- SOM: \$500M (Active CUDA migration targets in the next 18-24 months)

Competitive Advantage

- 5,000x ROI over human engineering hours

Competitive Landscape

Feature	Manual	ChatGPT	hipify-perl	RadeonShift
Syntax translation	Yes	Warn (hallucinates)	Yes	Yes
PTX risk detection	Yes	Warn	No	Yes
Wavefront64 optimization	Yes	Warn	No	Yes
Readiness scoring	No	No	No	Yes
CI/CD integration	No	No	No	Yes
Live MI300X telemetry	No	No	No	Yes